Silas Bambil

Statistics and Quantitative Risk

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FINRA Examinations

Series 63 – Uniform Securities Agent State Law Examination — Aug 2025
Securities Industry Essentials (SIE) — Jan 2025

Education

Georgia State University — Expected Spring 2028
MS in Mathematics, concentration in Statistics — *Atlanta, Georgia*

Georgia State University — May 2025
MS in Quantitative Risk Analysis and Management — *Atlanta, Georgia*

Kwame Nkrumah University of Science and Technology — Jul 2020
BSc in Actuarial Science — *Kumasi, Ghana*

Experience

Georgia State University — Aug 2026–Present
Graduate Lab Assistant — *Atlanta, Georgia*

– Provide instructional support in algebraic methods and problem solving for MATH 1111 and MATH 0999.

Osceola County School for the Arts — Jan–Jun 2026
AP Mathematics Teacher — *Osceola County, Florida*

– Taught AP Calculus, AP Precalculus, and AP Statistics; students achieved a 70% AP examination pass rate.

Georgia State University — Aug 2023–Jun 2025
Graduate Assistant — *Atlanta, Georgia*

– Consolidated graduate admissions records and constructed applicant-level predictors for enrollment modeling.
– Supported evaluation of the graduate-enrollment model and prepared statistical analyses for faculty research.

OPH CHEM CONSULT — May 2022–Aug 2023
Finance Associate — *Ghana*

– Examined financial misappropriation risks and community financial challenges; summarized findings in analytical reports and research briefs.

Ghana Statistical Service — Sep 2020–Aug 2021
Senior Researcher — *Ghana*

– Managed survey data and analyzed population and economic indicators for statistical reports.

Projects

Credit Risk Model Updating | R — 2026

– Compared retention, recalibration, and refitting across three Fannie Mae cohorts (about 1.55 million eligible loans), separating development, adaptation, and temporal test samples.
– Evaluated updating and regularization in 22,500 simulated datasets under five validation allocations; assessed performance on independent test samples.
– Examined how outcome-label budgets and distribution shifts affect the relative predictive performance of retention, recalibration, and refitting.

Fire Loss Modeling and Tail Diagnostics | R — 2026

– Analyzed about 1.25 million positive-loss NFIRS records (2016–2025); fitted generalized Pareto models to threshold exceedances to estimate Value-at-Risk and Expected Shortfall.
– Compared empirical and fitted expected payments within bounded loss layers; quantified uncertainty using incident-cluster bootstrap and assessed sensitivity to exceptional loss records.

shiftadapt | R Package — In development

– Implemented retention, intercept recalibration, intercept-and-slope recalibration, and full refitting.
– Separated fitting, validation, and test samples for minimum-log-loss and paired one-standard-error policy selection.

Publications

– Selecting and Estimating Credit-Risk Model Updates Under Distribution Shift. *Working manuscript*, 2026.
– Tail Model Diagnostics Using Expected Shortfall and Bounded Loss Layers with an Application to Fire Losses. *Working manuscript*, 2026.

Technical Skills

Software: R, Python, Microsoft Excel.
Statistical Methods: Logistic regression, Monte Carlo simulation, bootstrap, cross-validation, model calibration and validation.
Risk Modeling: Peaks-over-threshold (POT), generalized Pareto distribution (GPD), Value-at-Risk (VaR), Expected Shortfall (ES), scenario analysis.